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Cosmology intertwined: a review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies

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ABSTRACT

Review of 5.0 sigma tension Planck vs SH0ES, and S8 tension Planck vs weak lensing (KiDS, DES) and redshift surveys. Lists H0 evaluations from different teams/methods/geometric calibrations. New physics models: early dark energy, interacting dark sector, etc. Overview of next-decade experiments. Emphasizes fitting full data array, not single parameter.

What was actually measured (numbers):

5.0 sigma H0 tension, sigma8-S8 tension ~2-3 sigma, catalog of H0 values by team

Devices / Protocol:

Planck, SH0ES, H0LiCOW lensing, TRGB, BAO, RSD, Weak Lensing, Supernovae

Replication across teams / Independence:

Meta-analysis across dozens of independent probes; highlights persistence after years of analysis suggests not just systematics

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